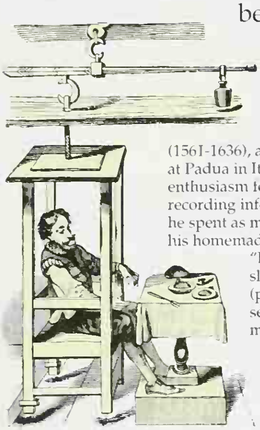
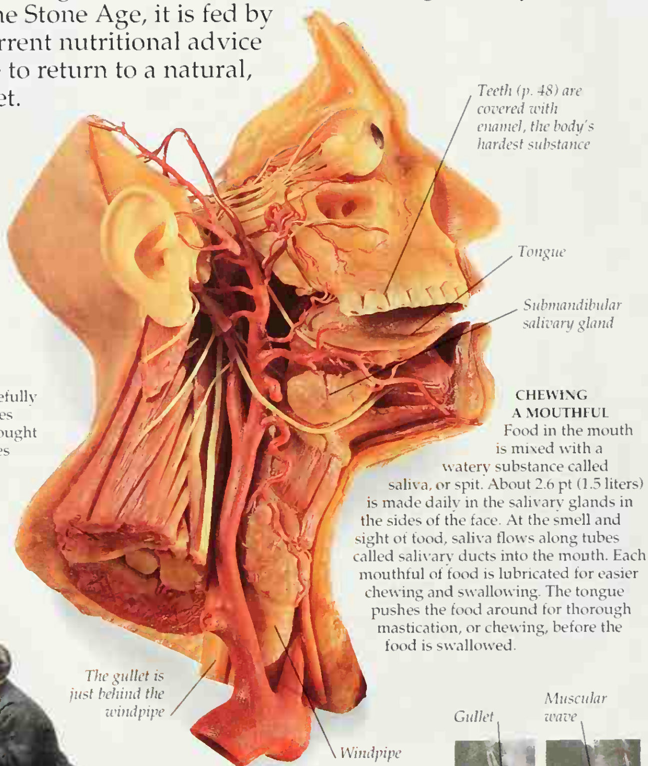
# Eating and digestion

A BALANCED DIET
There are six main nutrients in food needed for good health: carbohydrates (starches and sugars), fats, proteins, fibers, vitamins, and minerals. A traditional feast in many parts of the world has a different balance of food types to a modern Western meal. There is a good range of starchy foods like rice or potatoes for energy and fish for protein. Lots of fresh vegetables and fruits provide fiber, vitamins, and minerals. Few meat and dairy products are needed to provide sufficient protein and fats.

FOOD CONTAINS THE body's energy sources and the building blocks for maintenance and repair, besides being enjoyable. When you swallow, food travels along the digestive tract, from the mouth, down the gullet (esophagus) to the stomach, and then on to the small and large intestines. Because this tract is relatively large, it has been possible to study the mechanics of digestion since ancient times. However, detailed knowledge about the chemistry of digestion, such as why the body needs certain types of foods to maintain good health, is relatively recent. Even more recent is a drastic change, particularly in Europe and North America, from the age-old and varied human diet of fruits, vegetables, nuts, and some meats, to fast foods and processed foods that are rich in sugar, fat, and salt. So while the digestive system remains stuck in the Stone Age, it is fed by the Space Age. Current nutritional advice encourages people to return to a natural, better-balanced diet.



LIFE IN THE "BALLANCE"
Santorio Sanctorius (1561-1636), a medical professor at Padua in Italy, had a great enthusiasm for measuring and recording information. For 30 years he spent as much time as he could in his homemade weighing device, the "Ballance." Sanctorius ate, slept, defecated, excreted (pp. 38-39), and he even had sexual intercourse there, carefully measuring his weight changes after each activity. He thought unexplained weight losses might be caused by "invisible vapors" leaving the body.




CHEWING A MOUTHFUL
Food in the mouth is mixed with a watery substance called saliva, or spit. About 2.6 pt (1.5 liters) is made daily in the salivary glands in the sides of the face. At the smell and sight of food, saliva flows along tubes called salivary ducts into the mouth. Each mouthful of food is lubricated for easier chewing and swallowing. The tongue pushes the food around for thorough mastication, or chewing, before the food is swallowed.

CLAUDE BERNARD
French scientist Claude Bernard (1813-1878) is sometimes called the founder of experimental physiology – the story of the body's chemical workings. He found that only some stages of digestion take place in the stomach; the rest take place in the small intestine. Bernard also investigated the roles of the liver and pancreas and pioneered the principle of homeostasis, the idea that constant conditions are maintained inside the body.


X-RAYED SWALLOW
When a mouthful of chewed food is soft and pliable, the tongue squeezes off a portion. The food ball is pushed back and down into the throat, to the top of the gullet. Muscular waves in the gullet's wall propel the lump downward. In these X-rays the top half of the gullet is shown on the left and the bottom half on the right. The food contains barium, which shows up white.